

Crypto - My Why

Main Points

- enabling true digital ownership
 - this is valid for money but also (personal) data (eg verifiable credentials) as there can be a single source of truth against which can be verified
- openness and composability
 - accelerate innovation
 - once more and more services integrate blockchain things start to become "magically" composable
- reliability in the sense of
 1. always up computer
 2. programs written today will (most likely) keep functioning in the far future (hardness)

Properties

- verifiability (through re-execution or proof verification) allows for scalable self-governance
 - -> solves tragedy of the commons
- immutability: tamper-proof records of data history
- decentralized consensus: decide which reality is the real/canonical one
 - credibly neutral / open: censorship is not possible and anyone can use/join network
- resilience: a virtual computer that's always running
- hardness: how likely it is for something to be predictable/reliable in the future (ie commitment machine)
 - > missing (December 2025): privacy.

Crypto: Why are we here? | Erik Voorhees Keynote | Permissionless II

Crypto represents a rebellion against permission and coercion in finance. It offers individuals freedom to act without needing approval from authorities. This movement emphasizes the importance of building a fair financial system based on objective rules, rather than subjective laws.

- permissionless: novel, the ultimate goal, captures everything
- not freedom: permission to spend, to trade, ultimately to exist: implicit permission is given to you by banks, government when you use your bank account
- rebellion against permission/government:
 - free and sovereign individuals
- subversive degenerates, petulant teenagers who are too immature and naive to see the value of order?
 - won't bad actors thrive and society degenerate?

- **we don't seek freedom from permissions**
 - -> **transparent and objective rules** (which include access for everyone)
 - and not: opaque and subjective rules as right now
 - also mentioned in [Why Passive Investors Lose | Contrarian Investor Michael Green - Bankless | Podcast on Spotify](#) at the end:
 - more efficient and compliance built in
 - **code is law -> code is better than law**
 - rules of men: political process, fallible, room for interpretation
 - rules of code (physics, math): precise, infallible, respectable, universal
 - you don't care about rules or society. WRONG!
 - we do care!
 - **we are building a superior technology for rules within society**
 - we are building something meaningful, cherish it. wild and romantic, idealized
 - we build things in the open without forcing it on people
 - we have invented: world's first and only transparent, objective, permissionless financial system for all
 - capital will flow without permission into this decentralized financial system -> it is more efficient
-

Atoms, Institutions, Blockchains

- [Atoms, Institutions, Blockchains — Josh Stark](#)
- Hardness means making something likely to stay true in the future, and it comes from atoms, institutions, or blockchains. Atoms (physical limitations) and institutions (opaque, centralization of power, people) have limits, so blockchains offer a new, digital way to create hardness. Together, these sources help humans build trustworthy systems for law, money, and coordination.
- **hardness** -> how likely it is for something to be predictable/reliable in the future (ie commitment machine)
 - human civilization depends heavily on this bc it brings predictability to our civilization
 - most useful if it is customizable and programmable
 - 3 questions:
 - What is the hardness about? What is the “something” which is being made very likely to be true in the future?
 - What is the source of that hardness? What is the reason it is very likely to be true?
 - How hard is it? How can we measure the degree of hardness?
- sources of hardness:
 - **atoms**: physical stuff that has convenient properties (that are hard to change)
 - **institutions**: human created hardness that can be reliable and outlive single humans, exist for years, decades or even centuries.

- **blockchains:** new form of digital hardness created by cryptography, distributed computer networks and incentives
- Satoshi's invention was a new source of hardness. It was the first new source of hardness that humans have invented in thousands of years.
- Blockchains
 - credible neutral: transparent and verifiable rules
 - composability: applications can be very hard
 - hyperstructures: protocols that can run forever
 - blockchains are trustless in the sense that their hardness does not depend on people, but they are trustworthy in that they are still capable of producing hardness.
 - Ethereum enables web3: an ecosystem of web services, protocols, applications, and communities whose foundations are built primarily on blockchain hardness, rather than institutional hardness.
 - Blockchains cease being a novel contribution to humanity's hardness infrastructure when they collapse into just another institution. In other words, when they become centralized, dependant on specific groups of people.

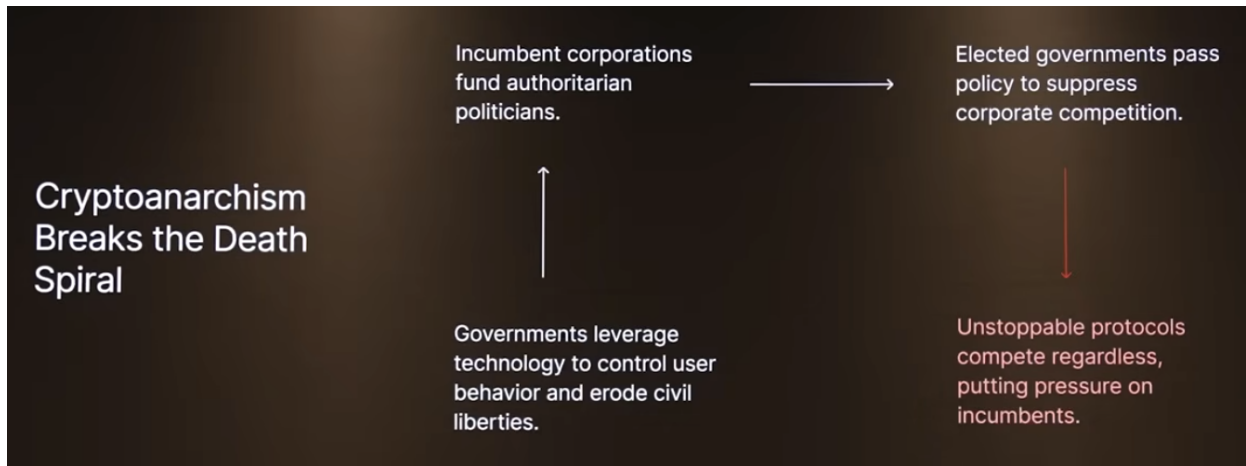
Read. Write. Own. - Chris Dixon

- enable true digital ownership of digital goods
- reliable (aka always up) and an app written today is guaranteed to work "as is" in the future (hardness)
- composability and open nature of ecosystem
- funding (tokens) is deeply ingrained in the network rules

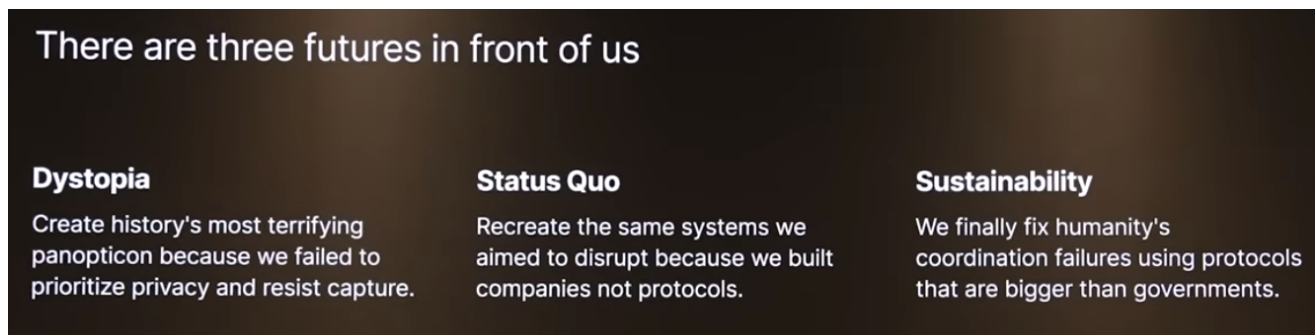
Why are we here? Crypto-Anarchism from First Principles)

- authoritarianism:
 - democracy is getting less over time
 - power is concentrating in a small group of companies (mostly tech)
 - almost unchecked power
- tragedy of the commons
 - incentives are not aligned -> a single farmer might have more sheep than a patch of grass can sustain but the cost is shared with all the other farmers
- governing the commons
 - centralization (communism)
 - privatization (capitalism)
 - self-governance
 - mutually enforced community governance. but how to scale without risk of corruption?
- anarchism:
 - not chaos, not to rules, it is no rulers

- simple philosophy that we should dismantle oppressive structures of power to improve human life
- a methodology not an end state
- crypto-anarchism
 - use cryptography and incentives to protect civil liberties and build coordination systems where enforcement happens peer-to-peer.
 - no specific watchmen/rulers -> everyone that participates (ie runs a node) can verify the rules
 - decentralization is anarchism
- crypto-anarchism breaks the death spiral



- decentralization accelerates the transition of goods to utilities
 - disrupt existing monopolies and technologies
 - institutions hate this
- **decentralization needs to be the means and not just the ends**
 - we need to build technologies that can't be stopped/censored



The purpose of smart contracts

- [# Lesson 1 Part 2 | The purpose of smart contracts - Solidity & Foundry Full Course](#)
- The world runs on agreements (contracts) enforced by law, culture, and trust.
- issue:
 - not immutable -> can be modified by parties in power
 - many times based on trust -> can be broken
 - or if law is violated it needs to go to costly court, which takes ages etc
 - -> better have agreements that can't be broken in the first place

- not verifiable -> oftentimes we need to trust a company or "name"
- not transparent -> we can't know what/how things are enforced within the agreement
- infrastructure guarantees trust-minimized interactions
 - verifiable: can read the code
 - transparent: executed according to protocol rules
 - credibly neutral: no one can be excluded
 -

Smart contracts

- **verifiable, trust-minimized agreements**
- **-> unbreakable promise**
- Smart contracts let us make **rules and agreements programmable** instead of relying on trusted middlemen (banks, companies, governments).
- Blockchains provide **trust, transparency, and immutability** through **decentralization** so code can enforce these rules automatically.
- brand based agreements -> math based agreements

Other benefits

- security
- uptime / resilience

Other thoughts

- bitcoin: why is it not going to zero despite security budget issues in the future?
https://x.com/drakejustin/status/1928025981270519924/?rw_tt_thread=True -> there's too many invested actors that worst case will start mining with a loss as they will otherwise lose even more money -> BTC might be too big to fail
- enables unbiased/neutral digital ownership and access -> has not been possible before

-
- decentralization as a core value
 - What are the properties of it?
 - Permissionless access
 - Self sovereign ownership
 - verifiability